

2015 Notes from the Marking Centre – Mathematics

General 2

Introduction

This document has been produced for the teachers and candidates of the Stage 6 Mathematics General 2 course. It contains comments on candidate responses to the 2015 Higher School Certificate examination, highlighting their strengths in particular parts of the examination and indicating where candidates need to improve.

This document should be read along with:

- the *Mathematics General 2 Stage 6 Syllabus*
- the 2015 Higher School Certificate Mathematics General 2 examination
- the marking guidelines
- Advice for HSC students about examinations
- other support documents developed by the Board of Studies, Teaching and Educational Standards NSW to assist in the teaching and learning of Mathematics in Stage 6.

Question 26

Candidates showed strength in these areas

- calculating required dosages for children given weight (part b)
- determining the relative frequency of an event from recorded data (part e)
- calculating the probability of the complement of an event (part e).

Candidates need to improve in these areas:

- calculating the distance, in kilometres, between two points on the same great circle (part c)
- calculating the surface area of a sphere (part f)
- calculating the cost of a phone bill given the details of the mobile phone plan (part g).

Question 27

Candidates showed strength in these areas:

- using scale factors to solve problems involving similar figures (part a)
- calculating drip rates and dosage strength (part b)
- determining the effect of additional data scores on the standard deviation (part d(ii)).

Candidates need to improve in these areas:

- solving simple linear simultaneous equations (part c)
- identifying the outlier in a data set using appropriate formula (part d(i))
- calculating the time to download a file (part e).

Question 28

Candidates showed strength in these areas:

- calculating the area of an annulus (part a)

- changing the subject of a formula and evaluating by substitution (part d)
- plotting ordered pairs of and estimating values from a line of fit by eye (part e).

Candidates need to improve in these areas:

- using calculated z-scores to compare scores from different data sets (part b)
- converting between common units for volume (part c)
- developing a linear equation from a description of a situation in which one quantity varies directly with another (part f)
- applying break-even analysis to a simple business problem (part f).

Question 29

Candidates showed strength in these areas:

- calculating measurements from a scale diagram (part c(i))
- calculating the total amount paid in a home loan situation (part b)
- calculating the mean from a table of values (part d(ii)).

Candidates need to improve in these areas:

- calculating credit card payments, interest charges and balances (part a)
- calculating the volume of a cylinder and applying algebraic skills to find a missing dimension (part c(ii))
- estimating the median from a cumulative frequency polygon (part d(i))
- interpreting and explaining the use of the graph of a quadratic function in a practical context (part e).

Question 30

Candidates showed strength in these areas:

- using a probability tree diagram to solve a problem involving a two stage event (part b)
- calculating the present value of an annuity using a table of present value interest factors (part c)
- applying the sine rule and cosine rule to find a side length and an angle of a triangle (part e(i) & part e(ii)).

Candidates need to improve in these areas:

- performing conversions involving units related to power (part a)
- calculating stopping distance (part d)
- applying trigonometry to solve a problem (part e(ii)).